

Extension of Glider Life using Structural Life Monitoring

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Abstract

The lifespan of a glider is limited by its susceptibility to fatigue damage. Repeated cyclic loading causes progressive damage accumulation, ultimately limiting its useful operational duration. The life limit currently imposed by EASA places a significant financial burden on glider clubs and contributes to composite waste, as no recycling solutions currently exist for end-of-life gliders.

This paper presents a Structural Life Monitoring (SLM) method to justify extended fatigue life, without revisiting certified structural limits or reopening Type Certificate documents. The method compares the EASA-recognized Kossira & Reinke reference load spectrum against the actual load spectrum measured during real flights, using the same Kossira & Reinke counting method for both, ensuring a consistent and legitimate comparison. An Equivalent Fatigue Index (EFI) is calculated using the Palmgren-Miner rule and conservative fatigue parameters. As long as the cumulative EFI remains below the certification reference limit, and the required inspection and continued-airworthiness provisions are satisfied, continued operation may be justified. The ratio between the reference EFI rate and the measured EFI rate quantifies the potential life extension.

Preliminary results on an instrumented glider show a ratio of around 26, corresponding to potential life extensions of several thousand hours for typical training and cross-country operations. This is consistent with life assessments performed on certain gliders and similar composite structures in other industries. The method is conservative by design, maintains the certified reference fatigue limit, and is compatible with EASA fatigue strength regulation requirements. A structured Concept of Operation and a certification pathway via EASA STC are proposed to support deployment.

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1. Introduction

Prediction of aircraft life is critical for ensuring safety and cost-effectiveness:

- Cost is affected by life because of design solutions, material selection, and associated manufacturing.
- Value is affected by life since used gliders depreciate suddenly when reaching their end of life.

This is becoming a costly dilemma for small glider clubs which may not be able to replace their old gliders with more recent new or used models.

The carbon footprint of composite structures is also a growing environmental concern since glider manufacturers currently do not provide solutions for recycling their composite gliders.

Extending glider life would extend their value and reduce the environmental impact.

Before getting into the methodology discussion, it is important to understand some key definitions: Service Life Limit, Reference Load Spectrum, Fatigue and EFI, S/N Curves and Safety Factor.

Service Life Limit: The Service Life Limit corresponds to the time a glider can fly, within the operational limits of its certified utilization category, without risking structural damage due to fatigue. Type Certificate holders may use different methods to demonstrate compliance with structural fatigue requirements, as accepted by the certification authorities. While these methods may vary, they all lead to an approved Service Life Limit. From an operational standpoint, it is this approved limit — not the method used to derive it — that governs how long a glider may remain in service. A typical glider Service Life Limit is 12,000 flight hours.

Utilization Category / Reference Load Spectrum: Gliders are certified to operate under a defined category. The Reference Load Spectrum corresponds to the load factor occurrences a glider may experience within its category over its entire Service Life. Reference Load Spectra have been established for the main glider categories, and are recognized by the certification authorities. Most gliders are certified under the Utility category, and the corresponding Kossira and Reinke spectrum [2] is recognized by the certification authorities.

Fatigue and Equivalent Fatigue Index (EFI): Aircraft structures are subject to variable loads throughout their service life, and repeated stress cycles can cause progressive structural deterioration — a phenomenon known as fatigue. Structural design aims to minimize stress concentrations, but fatigue accumulation must still be evaluated to ensure long-term structural integrity. One of the methods used in aerospace and accepted by EASA is the Palmgren-Miner rule, which combines load factor occurrences with the material's fatigue resistance. In the present SLM method, the cumulative Palmgren-Miner quantity is referred to as the **Equivalent Fatigue Index**, noted *EFI*. EFI is a conservative usage-severity index calculated consistently for both the certification reference spectrum and the measured operational spectrum.

S/N Curves (Wöhler Curves): Material fatigue resistance is characterized by the relationship between the stress amplitude applied to a specimen and the number of cycles it can sustain before failure. This relationship is represented graphically as an S/N curve, also known as a Wöhler curve, where S denotes the stress amplitude and N the corresponding number of cycles to failure. S/N curves are derived from laboratory testing. In the context of the Palmgren-Miner rule, the S/N curve is the link between the load spectrum and EFI calculation: for each stress level in the Reference Load Spectrum, the corresponding number of cycles to failure is read from the S/N curve, allowing the contribution of each load cycle to the cumulative EFI to be quantified. Due to design and operational limits, gliders only suffer from high cycle fatigue.

Safety Factor: Because uncertainties are inherent in the way structural fatigue and its associated limits are calculated, it is standard aerospace practice to apply an overall Safety Factor, even when margins are already included in individual calculations. The Safety Factor is applied to the Service Life Limit that is targeted. This means that fatigue calculations must be performed against a target equal to the intended Service Life Limit

multiplied by the Safety Factor. Aerospace typically uses a Safety Factor of 3. For example, a glider with a Service Life Limit of 12,000 flight hours must demonstrate structural integrity up to 36,000 hours of equivalent fatigue.

The load spectra, derived from Kossira & Reinke's seminal work [2], have become a key reference in defining load occurrence expectations for sailplane design and certification (EASA CS-22). These spectra were established using a compilation of a few hundred hours of real flight measurements performed on several gliders, flying in multiple conditions. The data and corresponding load factor occurrences have been extrapolated, using predefined operational profiles, to cover the envelope of all possible glider usages.

Recommendations by the European Union Aviation Safety Agency (EASA), as detailed in Certification Memorandum CM-S-006 [3], advocate for the use of these spectra for fatigue testing and structural analysis. In particular, the KoSMOS spectrum is the most recent version and includes omissions in the low load factor region, with no appreciable effect on damage calculation, to reduce testing time when actual laboratory fatigue tests are performed.

EASA and other authorities have accepted several methods for compliance with the fatigue strength regulatory requirement (CS 22.627), including full-scale fatigue testing based on an accepted reference spectrum as well as static load tests with additional required safety margins [4] [5] [6].

The granted service life limit, using either of the previously described methods, sets the level of maximum acceptable fatigue exposure. When using the Palmgren-Miner rule and a typical safety factor of 3, this means that the acceptable EFI limit should not exceed 1/3. If the approved service life limit is to remain meaningful, then the fatigue exposure associated with that limit must be treated as acceptable by definition. Otherwise, one would implicitly be stating that a glider could fail before reaching its approved life limit, which would contradict the purpose of the certification life limit. SLM does not challenge this approved limit; it uses it as the conservative reference against which measured usage is compared.

Attempts have been made in the past, using modern modeling tools and recently available material fatigue data, to quantify and justify that the composite structures accumulate damage at a much lower rate compared to what was expected during initial certification [7]. However, reopening existing Type Certificate documents to reflect these findings is impractical.

The proposed SLM concept is close to what is commonly called a model-based Structural Health Monitoring approach. In such an approach, the objective is not necessarily to identify the absolute physical damage state at every possible critical location. Instead, a conservative model of the ageing or fatigue mechanism is established, representative usage parameters are measured, and the measured usage is converted into a structural severity or fatigue exposure index. Continued-airworthiness decisions are then based on comparison between the measured index and the conservative reference model. Other SHM approaches exist, including statistical or neural-network-based methods, but they are not the basis of the proposed SLM concept. SLM is deliberately based on a transparent physical model, a recognized reference spectrum, and conservative assumptions, eliminating the need to reopen Type Certificate (TC) documents.

Similar principles have also been used in aviation usage monitoring and Individual Aircraft Tracking, particularly in military aviation where aircraft may experience missions significantly different from the original design assumptions [22] [23] [24]. The SLM proposal does not attempt to reproduce these military approaches directly, nor to claim that they are directly transferable to gliders. The relevant precedent is the general principle: measured usage can be compared with a conservative reference usage assumption to support individual aircraft life-management decisions, provided the method is controlled, conservative, and supported by inspections.

For gliders certified under CS-22, the goal is to compute the EFI associated with real measured usage using in-flight acceleration measurements. EFI can be computed using the Palmgren-Miner rule, the number of cycles to failure, the reference load spectrum, and conservative fatigue parameters. As long as the cumulative EFI stays below the certification reference limit, and the required inspection and continued-airworthiness provisions are satisfied, the operation of the glider can continue. The principle is shown in Figure 1.

Scope and limitations of the SLM method

The SLM method does not reopen the structural substantiation performed during Type Certification. Its objective is narrower: it uses the already accepted certified life exposure as a reference and compares it with the measured operational exposure of an individual glider. The Palmgren-Miner value is therefore used as an Equivalent Fatigue Index, calculated in the same way for the reference spectrum and for the measured spectrum.

This distinction is essential: SLM is a conservative reference-spectrum comparison method. Under the same calculation assumptions, the measured usage of an instrumented glider can be compared directly with the accepted certification reference usage, without requiring a new absolute fatigue-life prediction for the composite structure.

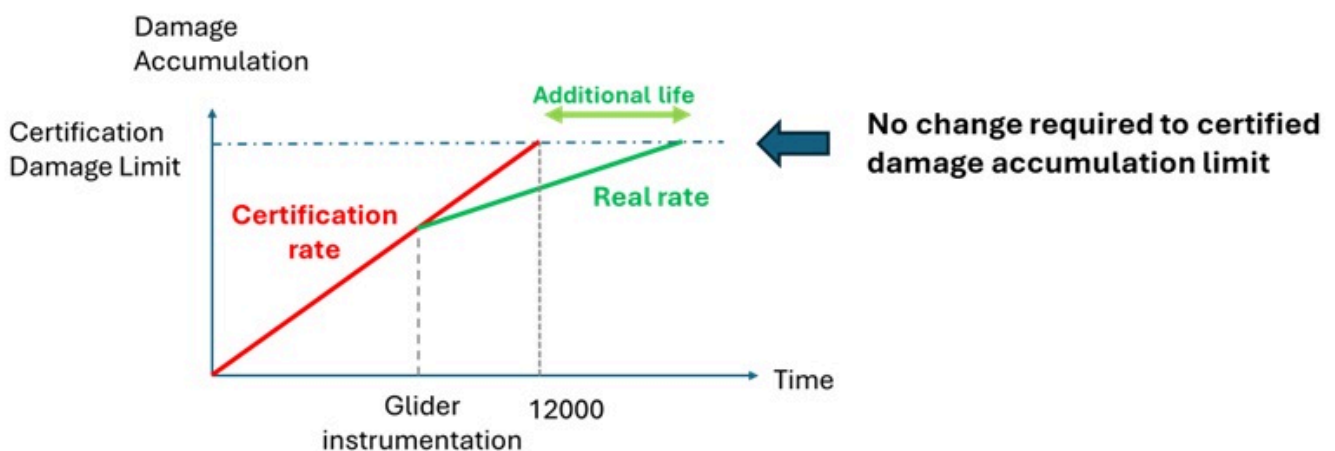


Figure 1: Life extension principle

2. Background and literature review

2.1 State Of the Art

2.1.1 Introduction of life limit in glider industry

Until the introduction of composite materials, gliders were manufactured using wood, fabric and aluminum. The fatigue behavior of these materials is well understood, only requiring standard practice to avoid any concentration of stress above certification limits. At the time of certification of the older gliders, no explicit fatigue life analysis or testing was required.

The first composite gliders were manufactured in the late 1950s: fs 24 Phönix first flight was in 1957, followed by Hütter H 30 GFK in 1962 (which led to the Glasflügel Libelle) and Akaflieg Darmstadt D-36 in 1964. The D-36 led to the ASW 12 (Waibel / Schleicher), the Cirrus (Holighaus / Schempp-Hirth) and LS1 (Lemke / LS) sailplanes which were then built in series. By the 1970s, composite materials became the standard in high-performance glider manufacturing, and all modern gliders today are made with advanced composite structures.

The glider community has been a pioneer in the use of composite materials, but industry's relative inexperience and limited understanding of fatigue behavior led some certification authorities to introduce a life limit. This 12,000-hour limit, which was set by LBA (Luftfahrt-Bundesamt), remains the standard life limit today under EASA.

It should be noted that, even if authorities like LBA imposed a relatively short limit on new designs, other authorities such as TRAFI [8] (Finland) and DGAC (France) approved much higher limits or even no limit (PIK20, Centrair C101).

2.1.2 Current state and possible evolutions

Today, the EASA has maintained the 12,000-hour life limit, although the CS 22.627 (Fatigue strength) only requires that "structure must be designed, as far as practicable, to avoid points of stress concentration where variable stresses above the fatigue limit are likely to occur in normal service".

In the glider industry, few full-scale fatigue tests have been performed. In most cases, the glider manufacturers have been complying with the requirement using the EASA accepted static overload test.

However, there has been one particularly interesting full-scale fatigue test performed in the 1990s, in Australia, on a Janus B wing [9]. The glider was not new and one wing had substantial repairs. Some defects were intentionally left unrepaired and some defects were created to simulate typical incidents (such as wing damage during out-landing). The conclusion was clear, confirming that there was no significant damage accumulation on the wing without repair and that only the non-repaired sections had substantial growth of minor damage. However, the propagation rate of delamination was slow and could be easily detected during the aging/fatigue inspection, which is mandatory at 6,000, 9,000, 10,000 & 11,000 hours.

The experience gained in the aerospace industry, as well as other industries like wind turbines, from inspection of aging composite structures, improved fiber characterization and better modeling, has validated the excellent fatigue behavior of composite airframes. This confirms that the historical 12,000-hour life limit imposed on gliders is extremely conservative as already demonstrated in a previous analysis "Numerical Comparison of Glider Load Spectra" [7].

Even if recent analyses have identified structural margin opportunities, it is not practical to reopen structure analysis for gliders certified 30 or more years ago. However, thanks to better understanding of glider operational usage and the corresponding composite-material EFI accumulation due to fatigue, our proposed method provides a solution to extend the life of gliders, without changing the certification damage accumulation limit associated with the safe life limit (12,000 hours).

2.2 Reference load spectrum, distribution occurrences and application to operational life assessment

The load spectra developed by Kossira & Reinke are foundational in sailplane design. In their seminal work, *"Determination of load spectra for the design of sailplanes"* [1], they established a methodology for deriving a reference load spectrum that includes both the stress levels and the occurrence frequency (distribution) of these load events.

To establish their load spectra, they instrumented gliders which were flown in different conditions, with the goal of being representative of the envelope of all possible flight usages. Because they could only fly for a limited time, it was necessary to scale the data to be representative of 6,000 or 12,000 flight hours required for certification. They used an extrapolation method which not only scales the occurrences as a function of time, but also increases the maximum and minimum load cycle values. While this extrapolation method is

understandable, from a statistical point of view to account for the lack of variability due to reduced flight time, it might be a source of over-estimation.

The counting method used by Kossira & Reinke is very similar to the "peak and valley" solution [10], with the addition of a simple filter to avoid counting minor load factor variations, and the optional use of a Markov transition matrix to count occurrences. These occurrences specify the expected number of cycles at various stress levels over a defined operational period, and form the basis for fatigue life predictions. However, it should be noted that these spectra have been developed to cover the envelope of all possible operations and have built-in conservatism compared to most typical operations.

The KoSMOS spectrum (Kollektiv für Segelflugzeuge, Motorflugzeuge und Motor-Segler, i.e. load collective for sailplanes, aeroplanes and powered gliders) is based on the original data from Kossira & Reinke with omissions of the low level stress variations. The goal is to significantly reduce test time when the spectrum is used for ground tests. Because our goal is to eventually compare the reference spectrum to the actual spectrum flown by a given glider, it is more suitable to use the original Kossira & Reinke spectra from their 1982 publication [2], which do not have these omissions.

Kossira & Reinke published multiple spectra corresponding to different utilization categories. The initial analysis presented in this paper uses the flight-only spectrum, which excludes simple aerobatics and ground operations. While most gliders are certified under the Utility category — which permits limited aerobatics — this conservative choice ensures that the results represent a lower bound of the potential life extension. Including simple aerobatics and ground operations would add occurrences at higher load factors, increasing the reference EFI accumulation and yielding a more favorable EFI ratio. Figure 2 is a representation of the Kossira & Reinke flight-only spectrum, normalized at 6,000 flight hours.

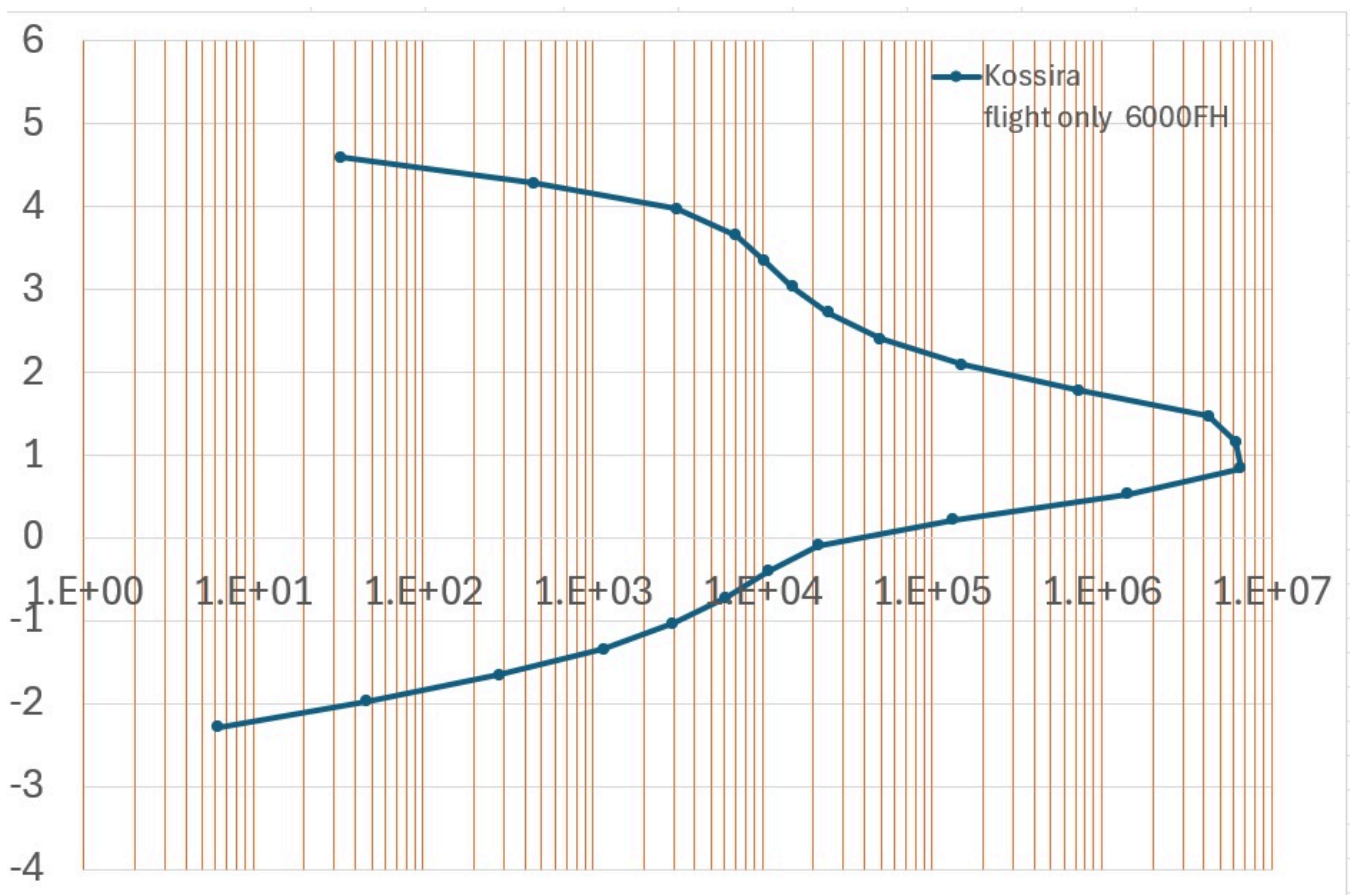


Figure 2: Kossira & Reinke flight-only spectrum normalized for 6,000 Flight Hours [2]
(x axis: occurrences / y axis: load factor)

Kossira & Reinke's subsequent work, *"Determination of load spectra and their application for keeping the operational life proof of sporting airplanes"* [2], extends these concepts to operational life assessment. The occurrence distribution, in this framework, enables the calculation of cumulative damage via cycle counts.

2.3 Kossira & Reinke counting method

Multiple counting methods have been developed over time, e.g. level crossing, peaks & valleys or Rainflow [10], and there are differences in the resulting occurrence calculations. The goal of this study is to compare the EFI accumulation between the reference spectrum and real flight spectra, and use the counting method developed by Kossira & Reinke [2] for their own spectra to compute occurrences for the real flight spectra.

2.3.1 Kossira & Reinke counting method principle

The Kossira & Reinke spectra were developed in the eighties to provide glider designers with load spectra which could be used for fatigue analysis, tests, and life justifications [2].

These spectra are referenced by EASA in their certification memorandum (CM-S-006 Issue 01 issued 20 January 2015) [3].

To establish these spectra, gliders have been instrumented to record the stress on the wing during multiple phases of flight.

Using the limited digital technology available at that time, the acquisition and processing were relatively simple:

1. Converting the wing loading, calculated with the spar bending moments divided by a reference bending moment, into digital classes of 2^{10} and 2^5 precision,
2. simple filtering to reduce the effect of small load factor variations (noise),
3. identifying maxima and minima in the smaller 2^5 class (maxima and minima are also called peaks & valleys),
4. filling a Markov transition matrix using "from stress/load factor" to "to stress/load factor" transitions which correspond to transitions from minimum (valley) to maximum (peak) as well as maximum (peak) to minimum (valley),
5. transforming the Markov matrix to count exceedances above and below the average load factor level,
6. summing the cells of the rows of the transformed Markov matrix to create the more user friendly Load Factor as function of occurrences exceedances graphic.

This final step provides the representation in Figure 2 and Figure 3 which are commonly called load spectra and which can be used for damage and life analysis. It is interesting to note that, while the values are slightly different, the shape and the average values between the reference and real spectra are consistent.

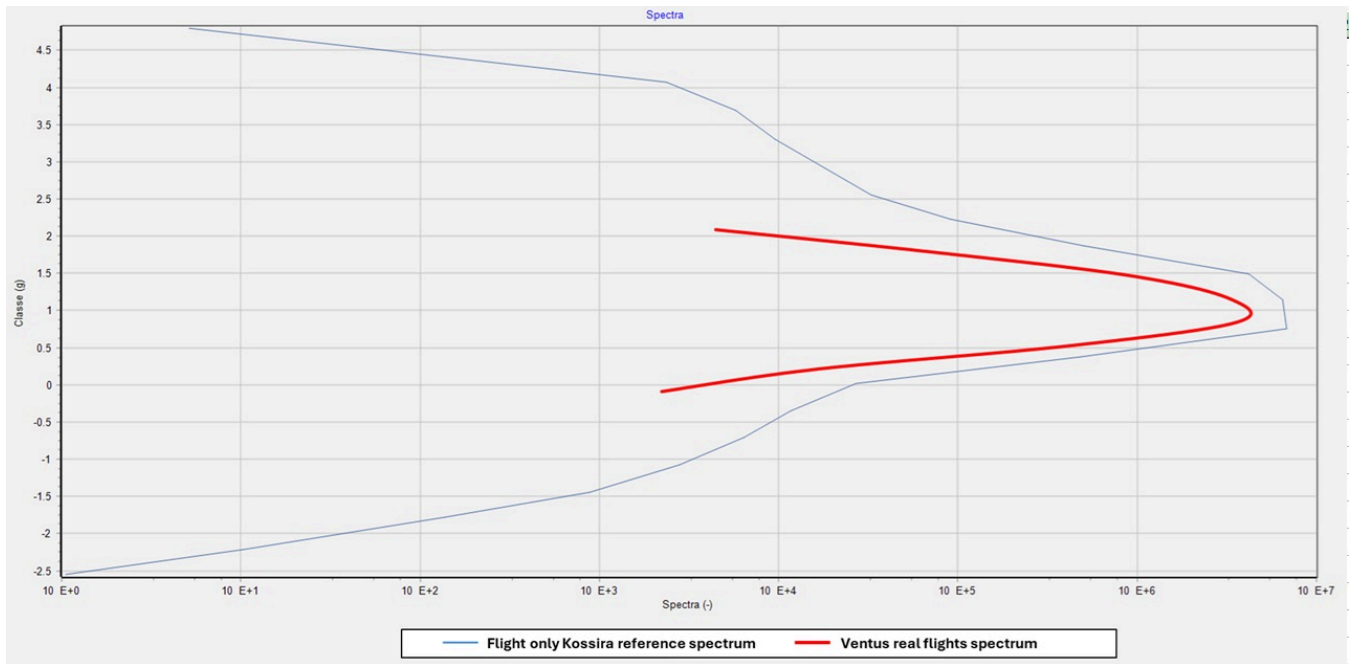


Figure 3: Examples of Kossira & Reinke flight-only spectrum and real flights Ventus spectrum
(x axis: occurrences / y axis: load factor)

2.3.2 Acquisition and classes

The sensor's signal, representing the glider body vertical axis load factor, is converted to digital values using 2^{10} precision (signal is quantified into 1024 classes). With a typical load factor range of -4g to +6g, the difference between two adjacent classes is small (approximately 0.01g). Analyzing the transitions using this accuracy results in a high number of small transitions, which have no real value for fatigue and life analysis. The solution proposed by Kossira & Reinke is to use a sub class using 2^5 precision (signal quantified in 32 classes). With such resolution, the difference between two adjacent 32-class values is approximately 0.32g which is more representative for fatigue and life analysis.

class #	class
	center (g)
	value
32	6
31	5.677419
30	5.354839
29	5.032258
28	4.709677
27	4.387097
26	4.064516
25	3.741935
24	3.419355
23	3.096774
22	2.774194
21	2.451613
20	2.129032
19	1.806452
18	1.483871
17	1.16129
16	0.83871
15	0.516129
14	0.193548
13	-0.12903
12	-0.45161
11	-0.77419
10	-1.09677
9	-1.41935
8	-1.74194
7	-2.06452
6	-2.3871
5	-2.70968
4	-3.03226
3	-3.35484
2	-3.67742
1	-4

Table 1: 32-Class definition

2.3.3 Filtering

Using data in the 32-class format is simple but has a negative effect since it amplifies small signal variations when these variations occur at the limit of the 32-class transitions. For example, with an input range of -4g to +6g, if the value of the input signal is 0.95g, the 1024-class value is 507 and the 32-class value is 16. If the signal increases slightly by e.g. 0.08g, the 1024-class value becomes 516 and the 32-class value 17. If the input signal oscillates between these values, the 32-class values also oscillate between 16 and 17, which means that a small 0.08g input signal variation is turned into a large 0.32g variation when converted to 32-class. To avoid these undesired 32-class variations induced by small input signal variations, Kossira & Reinke introduced a simple filtering solution by only taking into account a new sample, if the difference between this new sample and the last recorded sample exceeds a threshold DX , in the 1024-class. In their experimentations, Kossira & Reinke have used $DX=10$ [11], which means that the difference between previous sample and current sample has to be at least $\sim 0.1g$, for current sample to be processed. Figure 4 provides an illustration of the filtering/hysteresis proposed by Kossira & Reinke [2].

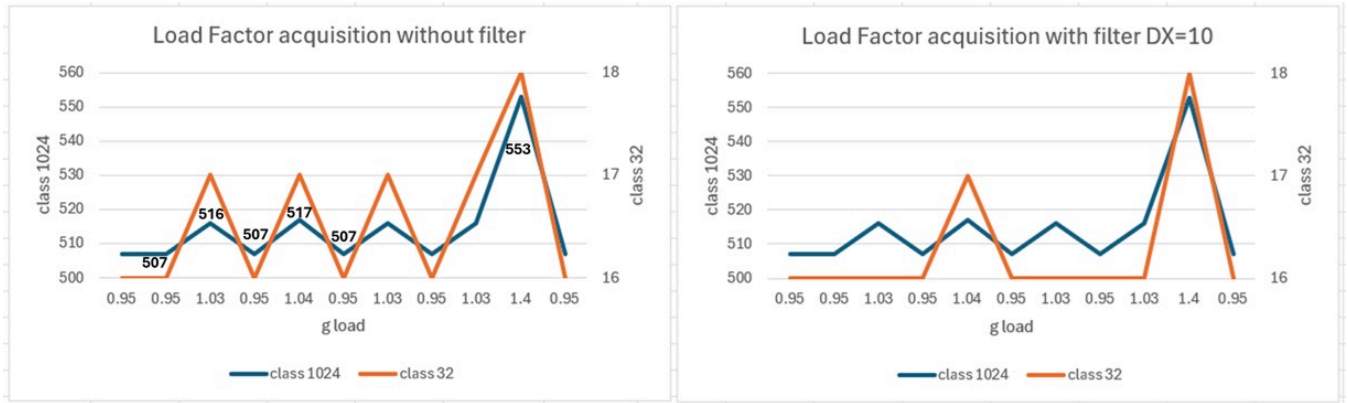


Figure 4: Kossira & Reinke filtering solution

2.3.4 Maxima and minima (peaks and valleys)

When the input data is filtered and reduced to the 32-class, it is straightforward to identify the minima and maxima in the data stream. The resulting data stream is a succession of minima and maxima in alternation, also called peaks and valleys, as shown in Figure 5.



Figure 5: Maxima & Minima / Peaks & Valleys

2.3.5 Markov transition matrix

The peaks and valleys data stream obtained from the previous steps is used to fill the Markov transition matrix. Because the data is in the 32-class format, the matrix size is 32 x 32. The columns of the matrix correspond to the "from" load factor of a transition and the rows of the matrix correspond to the "to" load factor of the transition. The peaks and valleys are processed by taking two consecutive values which respectively correspond to the "from" load factor and the "to" load factor. The corresponding cell of the Markov transition matrix is incremented. Figure 6 provides an example of a Markov matrix from a single flight.

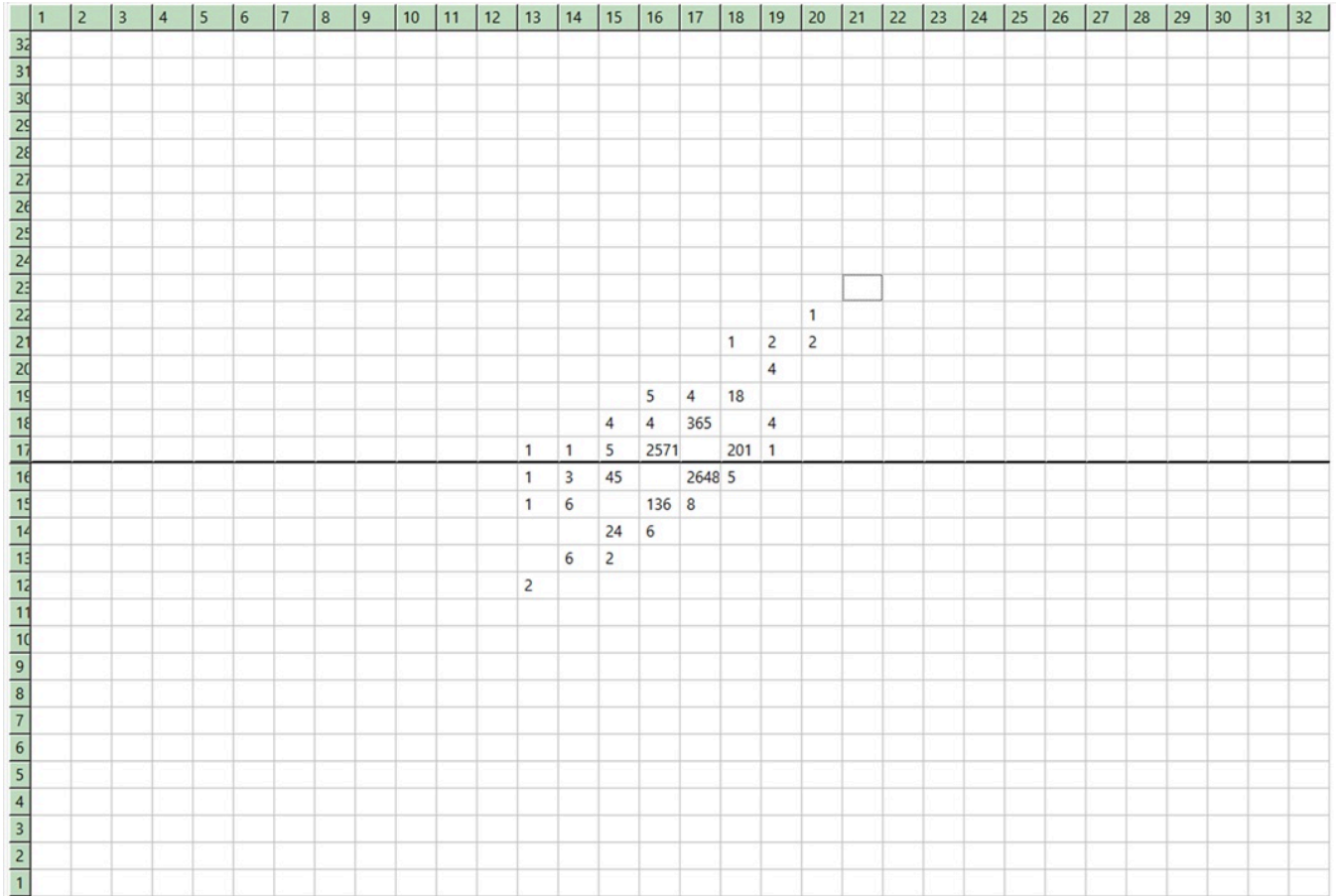


Figure 6: Example of Markov transition matrix
 (x axis: "**from**" level transition / y axis: "**to**" level transition)

2.3.6 Load factor as function of occurrences spectrum representation

To convert the Markov transition matrix data to the more traditional load factor as function of occurrences spectrum representation, it is necessary to first determine the average load factor value for the considered flight sequence(s).

Then compute how many transitions went through a given load factor above and below the mean value:

1. for each of the load factor cells above the mean value and above diagonal, compute the sum of all the occurrences at and above the considered cell (Figure 7 red/black example)
2. for each of the load factor cells below the mean value and below diagonal, compute the sum of all the occurrences at and below the considered cell (Figure 7 green/light gray example).

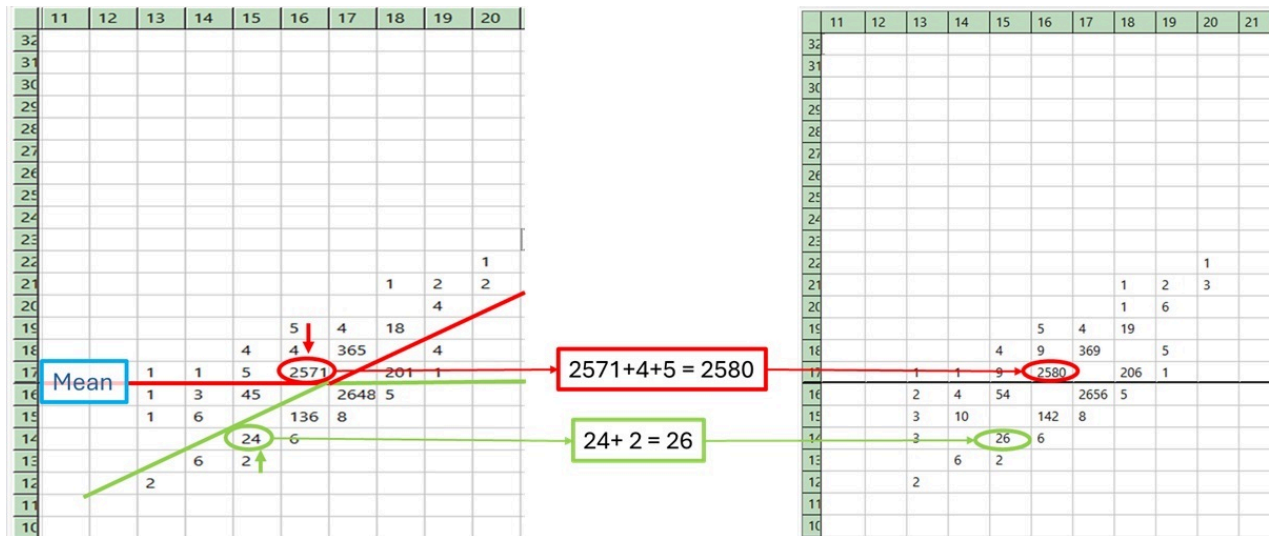


Figure 7: Calculation of the number of transitions through a given cell
(x axis: "**from**" level transition / y axis: "**to**" level transition)

For a given load factor level, sum all the occurrences to obtain the spectrum table and graphic representation, as seen in Figure 8 and Figure 9.

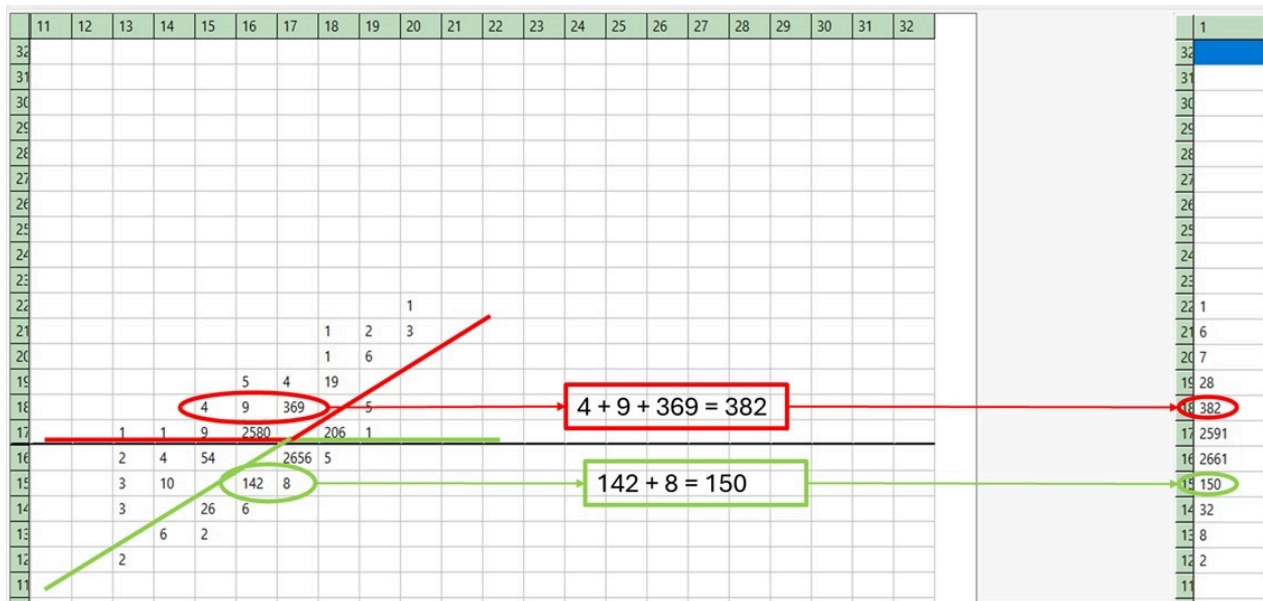


Figure 8: Calculation of the total number of occurrences for every load factor level
(Left = x axis: "**from**" level transition / y axis: "**to**" level transition
Right = x axis: number of occurrences / y axis: load factor)

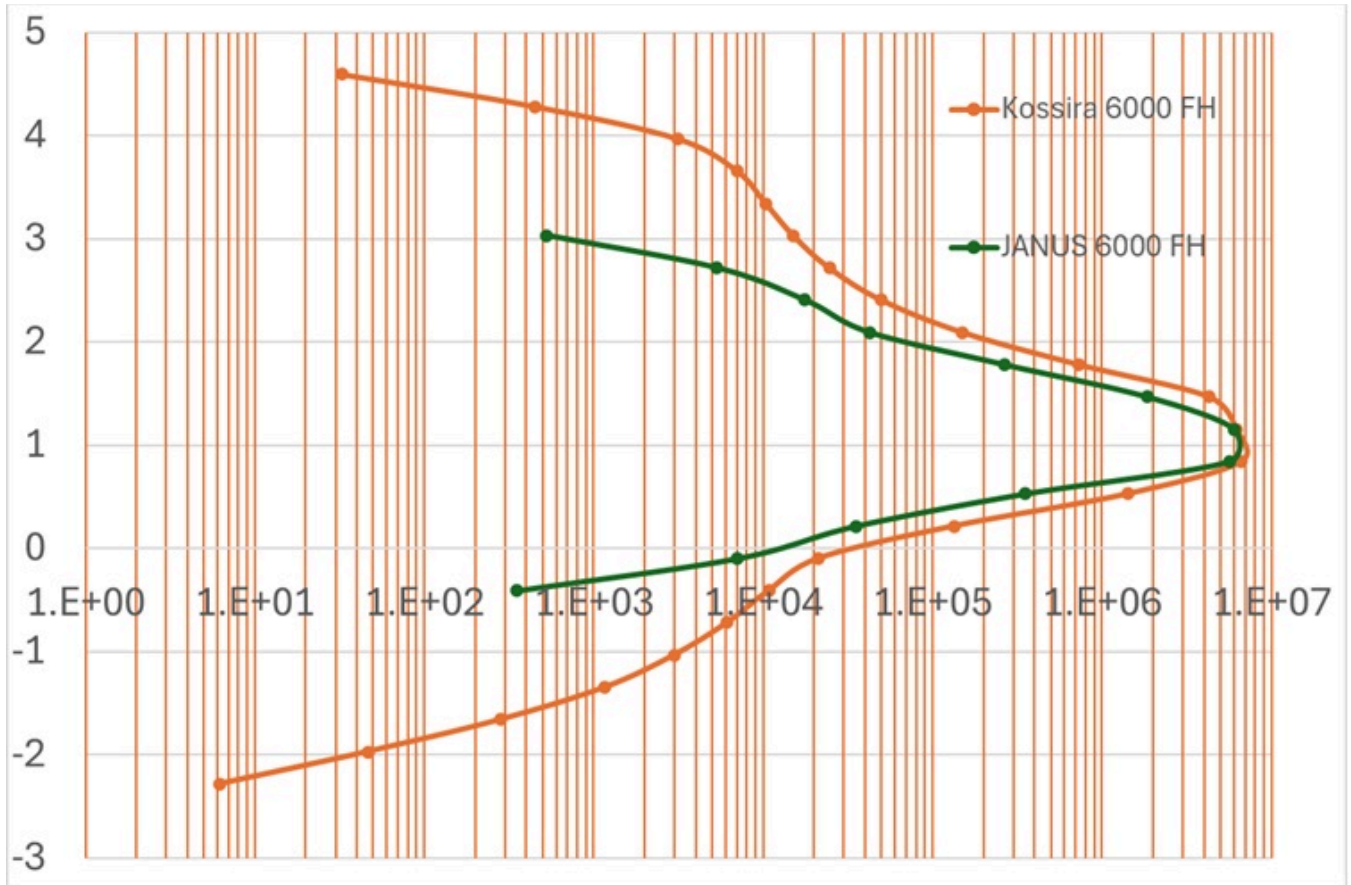


Figure 9: Example of load factor spectrum from a glider real flights in green/black with Kossira & Reinke reference flight-only (no aerobatics) spectrum in red/gray normalized at 6,000 flight hours

(horizontal axis: occurrences / vertical axis: g load)

2.4 Real load spectrum in glider operations

In actual glider operations, load conditions deviate from the conservative assumptions of reference models. Factors such as pilot technique, environmental variations, and operational profiles lead to a real load spectrum that often exhibits fewer or less severe load events than those predicted by the Kossira & Reinke distribution. The accelerations measured during actual flights are processed using the Kossira & Reinke counting method to provide these real load spectra.

2.5 Equivalent Fatigue Index: use of the Palmgren-Miner rule

Since the reference and measured spectra are established with the same counting method, it is reasonable to use the same calculation framework to compare their relative fatigue severity. The Palmgren-Miner rule [12] [13] provides a linear cumulative index for fatigue, expressed as:

$$EFI = \sum_i \frac{n_i}{N_i}$$

where n_i is the number of cycles at a specific stress level, and N_i is the number of cycles to failure at that level.

In classical Palmgren-Miner notation, this cumulative value is often denoted D , for damage. In the present SLM method, it is referred to as the **Equivalent Fatigue Index (EFI)**. EFI is a conservative usage-severity index calculated consistently for both the certification reference spectrum and the measured operational spectrum.

It is therefore possible to calculate an EFI from a given spectrum and compare it with a reference limit. In the present SLM application, the Palmgren-Miner rule is used as a common comparison framework between two spectra: the certification reference spectrum and the measured real-flight spectrum.

When multiple spectra are available, and the respective EFI values are estimated over the same exposure time, it becomes possible to directly compare them and estimate life extension when the measured real-flight spectrum is less severe than the reference spectrum.

2.6 Number of cycles to failure (N_i)

For a given load spectrum, either reference or from actual flight measurements, the occurrences at a given stress level n_i are known. To calculate the corresponding EFI for that spectrum, we only need N_i , which is the number of occurrences required to fail the structure at each load factor/stress level.

2.6.1 Material fatigue representation - S/N curve

Modeling fatigue is instrumental in understanding how materials behave under occurrences of loading. This is vital for industries such as aerospace, automotive, and construction. One tool used to represent fatigue data is the **S-N curve**, originally known as the Wöhler curve.

Wöhler's work has been at the root of modern fatigue analysis and testing. It introduced the first graphical representation of stress vs. cycles to failure and is now commonly referred to as the S-N curve as shown in Figure 10.

An S-N curve provides a representation of the relationship between the stress amplitude (**S**) applied to a material and the number of cycles to failure (**N**) it can endure.

- S : Stress a material experiences during a single cycle.
- N : Total number of cycles a material can withstand before failing.

For fatigue analysis and demonstrations, there is a direct correlation between stress cycles and load factor cycles due to the stress considered: structure flexion under load factor.

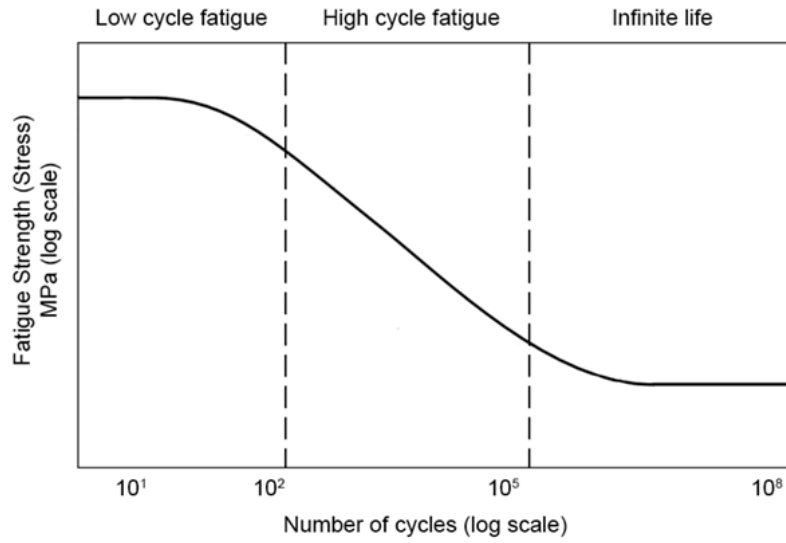


Figure 10: Typical Wöhler curve

2.6.2 Material fatigue representation - Basquin model

To be able to perform fatigue and life calculation, Basquin [14] proposed a simple mathematical model for the high-cycle fatigue region of the S/N curve. The standard stress-life form is:

$$\sigma_a = \sigma'_f (2N_f)^b$$

where:

- σ_a : stress amplitude
- σ'_f : fatigue strength coefficient, corresponding to the intercept of the elastic stress-life curve
- N_f : number of cycles to failure
- $2N_f$: number of stress reversals to failure
- b : fatigue strength exponent, i.e. the slope of the elastic stress-life curve in log-log representation

For the present application, all constants, including σ'_f and the factor 2^b , can be absorbed into a single coefficient C . The relation can therefore be written in the simplified form:

$$\sigma_a = C N^b$$

where N is the number of cycles to failure and C is a constant for the considered material system, laminate configuration and stress ratio.

This simplified Basquin power law expresses the stress amplitude σ_a required to produce failure in N cycles as a power-law function of N . It is appropriate for the high-cycle fatigue region of the S/N curve, which is the relevant regime for this application because, by design margins, a glider does not operate in low-cycle fatigue. At the other end of the curve, the Basquin model remains conservative because it does not credit any possible asymptotic endurance behaviour in the very low stress variation region.

2.6.3 End-of-life EFI reference normalization

By design, a glider has a finite fatigue life potential. While the glider is operated, this potential is consumed by the occurrence of load variations (cycles). The reference load factor spectrum and the measured real load factor spectrum are two examples of how the occurrences are distributed as a function of load factor and frequency.

Under the conventional Palmgren-Miner interpretation, the value $EFI = 1$ is associated with the theoretical failure threshold. In the SLM method, this same mathematical quantity is used as an equivalent fatigue exposure index.

Certification requirements and common practice incorporate a safety factor of 3, so that if a life of e.g., 12,000 hours is targeted, the calculations and validations should be performed for a duration of 36,000 hours.

Therefore, when using Kossira & Reinke as the reference spectrum, and Palmgren-Miner as a way to calculate the reference EFI, the normalized reference condition is:

$$EFI_{ref,36000} = \sum_i \frac{n_{i,ref,36000}}{N_i} = 1$$

when the spectrum of occurrences corresponds to 36,000 hours exposure time, and:

$$EFI_{ref,12000} = \sum_i \frac{n_{i,ref,12000}}{N_i} = \frac{1}{3}$$

when the spectrum of occurrences corresponds to 12,000-hour exposure time.

$$\boxed{EFI = \frac{1}{3}}$$

This EFI value corresponds to the 12,000-hour life potential limit for the SLM calculation.

This does not mean that the actual physical composite structure is assumed to fail at 36,000 hours, nor that its real physical damage state is known at 12,000 hours. In many composite fatigue tests, the structure may complete the required exposure without failure or with no significant observable damage. For SLM, the certification reference exposure is deliberately treated as if it consumed the full allowable fatigue budget. This is conservative because it does not take credit for any unobserved residual margin beyond the certified exposure.

2.6.4 Number of cycles to failure N_i from reference spectrum and fatigue parameters

Local stresses can be considered linear with the load factor around the average load factor:

$$\sigma_{a,i} \propto |LF_i - LF_{average}|$$

with LF_i the load factor and $LF_{average}$ the average load factor during the exposed time.

Using the Basquin model, with a constant C' which can be derived from the certification condition using the reference spectrum:

$$|LF_i - LF_{average}| \propto C' N_i^b$$

$$N_i = C' |LF_i - LF_{average}|^{1/b}$$

with C' being a proportionality constant which can be calculated using the reference EFI normalization:

$$EFI_{ref,12000} = \sum_i \frac{n_{i,ref,12000}}{N_i} = \frac{1}{3}$$

$$\text{with } N_i = C' |LF_i - LF_{average}|^{1/b}$$

$$\sum_i \frac{n_{i,ref,12000}}{N_i} = \frac{1}{3} \implies \sum_i \frac{n_{i,ref,12000}}{C' |LF_i - LF_{average}|^{1/b}} = \frac{1}{3}.$$

$$\sum_i \frac{n_{i,ref,12000} |LF_i - LF_{average}|^{-1/b}}{C'} = \frac{1}{3}.$$

$$\frac{1}{C'} \sum_i n_{i,ref,12000} |LF_i - LF_{average}|^{-1/b} = \frac{1}{3}$$

$$\implies C' = 3 \sum_i n_{i,ref,12000} |LF_i - LF_{average}|^{-1/b}.$$

The complete expression to calculate the N_i values which can be used to assess the cumulative EFI can be written as:

$$N_i = 3 \underbrace{\sum_j n_{j,ref,12000} |LF_j - LF_{average}|^{-1/b}}_{C'} |LF_i - LF_{average}|^{1/b}.$$

For a given spectrum n , LF and $LF_{average}$ are known, allowing calculation of C' and subsequently all the N_i values for each load factor of the spectrum.

These N_i values are only functions of the spectrum and the slope of the S/N curve b , as the Kossira & Reinke reference spectrum is known. If a conservative S/N slope b value is chosen to cover all types of composite glider structures, the resulting N_i values would become a reference usable for all EFI calculations.

3. Methodology

3.1 Data acquisition

3.1.1 Load cycle acquisition using accelerometers

From bending moment to load factor — establishing proportionality

To establish their spectra, Kossira & Reinke originally instrumented gliders with strain gauges on the wing spar at the root, measuring the spar bending moment M_{br} . A reference bending moment M_{br1g} was established by flying in calm air with a stable trajectory (vertical acceleration $\approx 1g$). Bending moments measured during flight were then divided by this reference value, yielding the normalized ratio $\frac{M_{br}}{M_{br1g}}$, which represents the wing

loading in flight.

When the glider is primarily subject to aerodynamic forces, this ratio is directly proportional to the vertical load factor N_z , which can be measured more practically using an accelerometer. This proportionality is the foundation of the SLM approach: if stress at any critical structural location is proportional to the load factor, then measuring load factor is sufficient to estimate EFI accumulation.

Validity of the proportionality within the certified flight envelope

One may argue that the relationship between load factor and local stress is not unique — that the maximum stress could occur at different locations on the structure depending on the aerodynamic and mass distribution along the wing, and that the coefficient of proportionality between stress and load factor is therefore unknown.

This objection is valid in general, but does not apply when the glider is operated within its certified flight envelope. By design, the manufacturer has ensured that, for any combination of speed, flap and airbrake configuration, and mass loading within the authorized range — covering fuselage weight (pilots) and wing mass (including water ballast) — the resulting stress at all critical structural locations remains below the maximum allowable. The structural design therefore implicitly bounds the stress at every critical point as a function of load factor, within the operational envelope.

This is further reinforced by the use of load factor limits in glider operational documentation. The maximum and minimum load factor limits published in the Flight Manual are precisely the operational expression of this design constraint: they are the values beyond which structural stress would exceed acceptable levels. As long as the glider is flown within these limits and within the authorized configuration and mass envelope, load factor is a valid and sufficient proxy for structural stress.

Furthermore, it is important to note that the absolute value of the proportionality coefficient between load factor and local stress is not required. The Palmgren-Miner EFI calculation relies on stress amplitudes — that is, the variation of stress between successive peaks and valleys — rather than absolute stress values. Since stress is proportional to load factor within the authorized flight envelope, stress amplitude is equally proportional to load factor amplitude. The proportionality coefficient therefore cancels out when computing the EFI ratio between the reference and real spectra. What matters is not the magnitude of the coefficient, nor where on the structure the maximum stress occurs, but simply that the glider is operated within its authorized flight envelope, ensuring that load factor variations remain a consistent and valid representation of stress variations throughout the structure.

Although not explicitly documented in their work [11], this is probably the reason why Kossira & Reinke only instrumented the glider under test at the root of the wing spar.

They were looking for stress variations centered around the average flight stress (i.e. $\frac{M_{br}}{M_{br1g}}$).

Accelerometers as a practical substitute for strain gauges

This equivalence makes accelerometer-based recording a cost-effective and practical substitute for strain gauges in the context of SLM. The accelerometer should be located as close as possible to the Center of Gravity (CG) to minimize contamination from centripetal accelerations during attitude changes. Practically, a location in the rigid structure close to wing spar intersection, within the fuselage is appropriate.

Non-flight phases

The proportionality between load factor and spar stress holds during normal flight phases. During ground rolls or winch launches, significant non-aerodynamic forces are present and the relationship no longer holds directly. Until dedicated flight physics models are validated for these phases, a conservative fixed contribution to the fatigue spectrum can be added, based on the same assumptions Kossira & Reinke applied for ground maneuvers and winch launches. The approach adopted for the initial experimentation is described in section 3.1.2.

3.1.2 Data acquisition for experimentation

Real flight load factor N_z is recorded in real time using accelerometers located on the wing spar (close enough from CG and very rigid), and electronic memory (e.g. SD card).

This data is processed using the Kossira & Reinke method to build the real flight load spectra. Using the same method as Kossira & Reinke for their reference spectra is important to make comparison between real and reference spectra legitimate.

During the initial experimentation, the takeoff and landing roll are treated as normal flight sequences. This is conservative since the Kossira & Reinke flight-only reference spectrum does not include ground operation phases. Any load factor occurrences recorded during takeoff and landing rolls are added to the real spectrum without a corresponding contribution in the reference spectrum, resulting in a lower Palmgren-Miner ratio at those load factor levels, and therefore a more conservative overall life extension estimate. No winch takeoffs are being planned with the experimentation gliders but, if this was the case, the same Kossira & Reinke assumptions would be used to add the corresponding load cycle occurrences to the experimentation glider spectrum. This is believed to cover for the limitations of using accelerometers instead of strain gauges.

Since most of the energy from turbulence (CS 22.341) or glider movement is below 10 Hz, the accelerometers should be filtered with a cutoff of at least 10 Hz, requiring a minimum sampling rate of 20 Hz. Accelerometer calibration can be checked before every flight when the glider is static to make sure the magnitude of the gravity measured on the 3 axes is $\sim 1g$.

3.2 Load spectrum analysis

Two load spectra are defined:

- **Reference Load Spectrum:** Based on Kossira & Reinke (as recommended by the European Union Aviation Safety Agency), this reference spectrum includes the stress levels (load factor) and their corresponding distribution occurrences (Figure 2).
- **Real Load Spectrum:** Derived from flight data, representing the actual occurrences of various stress levels during operations. To be consistent with the reference spectrum, the Kossira & Reinke counting method (§ 2.3) shall be used.
Within the scope of the initial experimentation, all phases of flight were included (takeoff, flight and landing).

To allow comparison between spectra, the same load factor range should be used, and the respective cycle counts n_i should be normalized to the same flight duration.

3.3 Application of the Palmgren-Miner rule to estimate EFI and life extension

3.3.1 EFI condition to continue operation beyond the 12,000-hour limit

Since the measured spectrum is known, and since the same conservative fatigue model is applied to both the reference and measured spectra, it is possible to calculate the EFI associated with the measured usage.

The operational condition for continued use is an inequality: the cumulative EFI must remain below the certification reference limit:

$$EFI_{total} \leq \frac{1}{3}$$

The equality $EFI = \frac{1}{3}$ is used only to calibrate the reference calculation. As explained in Section 2.6.3, the approved 12,000-hour life limit is treated as if it exactly consumed the allowable fatigue budget, even though the real structure may in practice have consumed less fatigue potential and may have passed the substantiation exposure without failure or significant observable damage. Since the real residual margin is not known and is not credited, the SLM method conservatively normalizes the reference case as:

$$EFI_{ref,12000} = \frac{1}{3}$$

This is a conservative hypothesis: the certified reference exposure defines the allowable EFI budget, and no additional unobserved margin beyond that approved exposure is used to justify continued operation.

Using this conservative normalization, the number of cycles to failure used in the EFI calculation is:

$$N_i = C' |LF_i - LF_{average,real}|^{1/b}$$

and the measured real-flight EFI is:

$$EFI_{real} = \frac{1}{C'} \sum_i \frac{n_{i,real}}{|LF_i - LF_{average,real}|^{1/b}}$$

where:

$$C' = 3 \sum_i n_{i,ref,12000} |LF_i - LF_{average,ref}|^{-1/b}$$

As long as the total EFI remains below the reference certification limit, and provided the required inspections and continued-airworthiness conditions are satisfied, the glider may continue operation.

In reality, gliders will only be instrumented late in their life. The total EFI will be a combination of reference EFI for the period when the glider was not instrumented and measured EFI for the period when the glider is instrumented:

$$EFI_{total} = EFI_{glider \text{ not instrumented}} + EFI_{glider \text{ instrumented}}$$

where $EFI_{glider \text{ not instrumented}}$ is computed using the reference spectrum normalized for a duration equal to the time during which the glider was flying without instrumentation,
and where $EFI_{glider \text{ instrumented}} = EFI_{real}$ is computed using the real flight occurrences. There is no need to normalize as it corresponds to the duration when the glider was instrumented.
The condition is that the glider can stay in operation as long as $EFI_{total} \leq \frac{1}{3}$ and the associated inspection and continued-airworthiness conditions are satisfied.

3.3.2 Life extension estimation

Once the EFI from actual flights has been computed using the Palmgren-Miner framework, it is possible to estimate how much longer a glider can operate beyond the certified 12,000-hour life limit, with the important assumption that future usage remains similar to the measured flights already collected.

3.3.2.1 EFI accumulation rate

As EFI accumulation is assumed to be linear with respect to time under a given load spectrum, the total EFI accumulated over time can be expressed as:

$$EFI = t \cdot EFI_{unit},$$

where:

EFI is the total accumulated equivalent fatigue index,

t is the flight time,

EFI_{unit} is the EFI per hour under a specific load spectrum.

3.3.2.2 Remaining real life and Palmgren-Miner ratio

Let:

$EFI_{unit,real}$ = EFI accumulated from the measured real-flight load spectrum in 1 hour

$EFI_{unit,ref}$ = EFI predicted by the certified Kossira & Reinke reference load spectrum in 1 hour

$t_{remaining \text{ ref}}$ = flight time remaining until the certified 12,000 hours are reached under reference spectrum usage

$t_{remaining \text{ real}}$ = estimated remaining life under measured usage

$EFI_{remaining}$ = remaining EFI potential until the certification limit $EFI = \frac{1}{3}$ is reached

For a given remaining EFI potential $EFI_{remaining}$, the time required to consume it differs based on the spectrum used:

- Under the certified reference spectrum:

$$t_{remaining \text{ ref}} = \frac{EFI_{remaining}}{EFI_{unit,ref}}$$

- Under the measured real-flight spectrum:

$$t_{\text{remaining real}} = \frac{EFI_{\text{remaining}}}{EFI_{\text{unit,real}}}$$

The Palmgren-Miner ratio R is therefore defined as the ratio between the reference EFI rate and the measured real-flight EFI rate:

$$R = \frac{EFI_{\text{unit,ref}}}{EFI_{\text{unit,real}}}$$

Substituting into the previous expression:

$$t_{\text{remaining real}} = t_{\text{remaining ref}} \times R$$

Thus, if measured real-world flights result in a significantly lower EFI rate than the conservative certified reference assumption, the remaining life is proportionally extended.

3.3.2.3 Additional life

The additional flying time enabled by lower real-world fatigue loads is:

$$t_{\text{extension}} = t_{\text{remaining real}} - t_{\text{remaining ref}}$$

$$t_{\text{extension}} = (R - 1) \times t_{\text{remaining ref}}$$

To ensure the Palmgren-Miner ratio R is valid, both EFI values must be computed over the same duration (e.g., the same number of flight hours).

For the present application, R is estimated using the measured real spectrum, which has a duration of t_{real} , and the reference spectrum normalized to the same t_{real} hours:

- t_{real} = duration of real flight data collected (in hours)
- EFI_{real} = EFI accumulated from the measured real-flight load spectrum in t_{real} hours
- EFI_{ref} = EFI predicted by the certified Kossira & Reinke reference load spectrum in t_{real} hours

$$\begin{aligned} EFI_{\text{ref}} &= EFI_{\text{unit,ref}} \times t_{\text{real}} \\ EFI_{\text{real}} &= EFI_{\text{unit,real}} \times t_{\text{real}} \\ R &= \frac{EFI_{\text{ref}}}{EFI_{\text{real}}} \end{aligned}$$

If the reference spectrum has been calculated over the typical 6,000 flight hours, for R calculation, it should be normalized to the time used for real spectrum observation t_{real} , using $\lfloor \frac{n_{i,\text{ref}} \times t_{\text{real}}}{6000} \rfloor$. The purpose of using the lower integer part of a real number (denoted $\lfloor \cdot \rfloor$) is to provide additional conservatism when adjusting the n_i of the reference spectrum to the same exposure time as the real flight data, as the latter is much smaller.

Therefore it is possible to calculate R :

$$R = \frac{\sum_i \lfloor \frac{n_{i,\text{ref}} \times t_{\text{real}}}{6000} \rfloor \times |LF_i - LF_{\text{average,ref}}|^{-1/b}}{\sum_i n_{i,\text{real}} \times |LF_i - LF_{\text{average,real}}|^{-1/b}}$$

Where:

- $n_{i,ref}$ and $n_{i,real}$ are the number of occurrences at each load factor level LF_i in the reference and real spectra, respectively.
- $LF_{average}$ is the mean load factor for the respective spectrum.
- b is the slope of the S/N curve.

$$t_{\text{extension}} = t_{\text{remaining ref}} \times \left(\frac{\sum_i \left\lfloor \frac{n_{i,ref} \times t_{real}}{6000} \right\rfloor \times |LF_i - LF_{\text{average,ref}}|^{-1/b}}{\sum_i n_{i,real} \times |LF_i - LF_{\text{average,real}}|^{-1/b}} - 1 \right)$$

To illustrate the time extension estimation, if the measured real-flight EFI over 25 hours is 0.000035, and the reference EFI over the same duration is 0.0007 (i.e., $R = 20$), and the glider has 500 hours left before 12,000 hours under reference spectrum use, the extension is $t_{\text{extension}} = 500 \times (20 - 1) = 9,500$ hours.

As the following paragraphs will show, despite the variability from glider to glider, due to airframe response to turbulence, type of operations, as well as pilot skills, the R factor can reach a few hundred. To ensure significant life extension is achieved, especially when glider operations result in a lower R value, it is recommended to start instrumentation as soon as possible. As an example, it might be wise to instrument the glider when the last 1,000-hour check is performed, before reaching the 12,000-hour limit.

However, it should be understood that the R ratio and time extension are highly dependent on the way the glider is being flown. Gliders being flown aggressively, performing spin entry/recovery or other maneuvers requiring a wide load factor range, will have much less favorable time extension than gliders being flown cross-country with pilots flying softly. In essence, regardless of the mission profile, gentler flying techniques will result in a greater potential life extension.

3.3.3 Material parameter b

Once the reference and measured spectra are defined, the parameter b of the Basquin equation, i.e. the S/N curve slope, is the only remaining parameter affecting the EFI ratio calculation.

In the SLM method, the selected parameter b is used as a conservative weighting function to compare the relative severity of two load spectra: the accepted certification reference spectrum and the measured operational spectrum. The same fatigue model and the same value of b are applied to both spectra, so the primary result is the ratio between the reference EFI rate and the measured EFI rate.

The choice of b remains important because it controls the relative penalty applied to high-load events. A steeper conservative value gives more weight to occasional high-load cycles and therefore reduces the calculated life-extension ratio when the glider is flown aggressively. This is desirable for SLM because it makes the method usage-sensitive and penalizes precisely the type of operation that is more fatigue-relevant.

Because this parameter is important to the complete demonstration, it should not be justified only from general composite fatigue textbooks. It should be selected from explicit fatigue fits obtained on composite structures that are reasonably close to glider structures in terms of material family, high-cycle fatigue regime and large lightly loaded composite construction.

The present SLM method targets ageing composite glider primary structures made predominantly from conventional GFRP/CFRP epoxy laminates. Therefore, the Basquin parameter shall be selected primarily from epoxy-based GFRP/CFRP fatigue data. Vinyl ester and polyester glass-fibre data are not considered representative of composite glider primary structures and are not used to define the glider parameter. They are nevertheless retained as a secondary comparison because they represent lower-performance

glass/polymer systems and provide a useful check that the selected value remains conservative for a broader family of composite structures.

For this purpose, wind turbine blade composite literature is particularly useful. Wind turbine blades are not aircraft structures, but they are large GFRP/CFRP composite structures operating in high-cycle fatigue under variable-amplitude spectra. The SNL/MSU/DOE fatigue database and associated Sandia reports provide explicit fatigue fit parameters for a large number of wind-blade-relevant composite laminates, adhesives and structural details [15] [16] [17] [18]. These references use the power-law form:

$$S = AN^B$$

where S is maximum applied stress or strain, N is cycles to failure, and A and B are fitting parameters. The stress-fit exponent B is directly equivalent to the Basquin exponent b used in this paper:

$$b = B$$

and the exponent used in the EFI equation is:

$$K = -\frac{1}{b} = -\frac{1}{B}$$

The following values are extracted from SNL/MSU/DOE wind-blade composite fatigue data. They are separated into two tables to avoid mixing the primary evidence used for the selected glider parameter with secondary non-epoxy comparison data.

The first table is the primary basis for the parameter selection because it corresponds to epoxy-based GFRP/CFRP laminates, which are the closest available published analogue to composite glider structures.

Source	Material / detail	Matrix	Loading			Relevance
SAND2010-7052, Table 9 [15]	QQ1 glass/epoxy multidirectional laminate	Epoxy	$R = 0.1$	-0.1556	6.43	Steepest identified GFRP epoxy value; primary basis
SAND2010-7052, Table 9 [15]	QQ1 glass/epoxy multidirectional laminate	Epoxy	$R = -1$	-0.1378	7.26	Reversed loading case; primary comparison
SAND2010-7052, Table 9 [15]	QQ1 glass/epoxy multidirectional laminate	Epoxy	$R = -0.5$	-0.1407	7.11	Mixed tension-compression case; primary comparison
SAND2010-7052, Table 9 [15]	QQ1 glass/epoxy multidirectional laminate	Epoxy	$R = 0.5$	-0.1313	7.62	Tension-tension case; primary comparison
SAND2010-7052, Table 9 [15]	QQ2 glass/epoxy multidirectional laminate	Epoxy	$R = 0.1$	-0.1216	8.22	GFRP epoxy laminate; primary comparison
SAND2010-7052, Table 9 [15]	QQ4 glass/epoxy multidirectional laminate	Epoxy	$R = 0.1$	-0.1263	7.92	GFRP epoxy laminate; primary comparison
SAND2010-7052, Table 9 [15]	TT-TPI-EP glass/epoxy blade laminate	Epoxy	$R = 0.1$	-0.1208	8.28	GFRP epoxy blade laminate; primary comparison
SAND2010-7052, Table 9 [15]	TT VARTM glass/epoxy laminate	Epoxy	$R = 0.1$	-0.1327	7.54	Infused GFRP epoxy laminate; primary comparison

Source	Material / detail	Matrix	Loading	$B = b$	$K = -1/b$	Relevance
AIAA 2012 / SNL-MSU-DOE database, Table 7 [16]	Multidirectional glass/epoxy laminate $[(\pm 45)_2/(0)_2]_s$	Epoxy	$R = 0.1$	-0.130	7.69	GFRP epoxy multidirectional laminate; primary comparison
AIAA 2012 / SNL-MSU-DOE database, Table 7 [16]	Triax glass/epoxy laminate $(\pm 45/0)_s$	Epoxy	$R = 0.1$	-0.143	6.99	GFRP epoxy triax laminate; primary comparison
AIAA 2012 / SNL-MSU-DOE database, Table 7 [16]	2-ply-drop delamination detail	Epoxy detail	$R = 0.1$	-0.120	8.3	Structural-detail fatigue; primary comparison
SAND2010-7052, Table 9 [15]	P2B carbon/glass hybrid, carbon-dominated axial laminate	Epoxy	$R = 0.1$	-0.0226	44.25	CFRP-like behaviour; much less fatigue-sensitive
SAND2010-7052, Table 9 [15]	P2B carbon/glass hybrid, carbon-dominated axial laminate	Epoxy	$R = -1$	-0.0379	26.39	CFRP-like reversed loading; much less fatigue-sensitive
SAND2010-7052, Table 9 [15]	P2B carbon/glass hybrid, carbon-dominated axial laminate	Epoxy	$R = -2$	-0.0455	21.98	CFRP-like mixed loading; much less fatigue-sensitive

The second table is retained only as secondary comparison. Vinyl ester and polyester systems are not considered representative of composite glider epoxy primary structures and are not used to define the selected glider parameter.

Source	Material / detail	Matrix	Loading	$B = b$	$K = -1/b$	Relevance
SAND2010-7052, Table 9 [15]	TT-TPI-VE glass/vinyl ester blade laminate	Vinyl ester	$R = 0.1$	-0.1384	7.23	Secondary only; not representative of glider epoxy structures
SAND2010-7052, Table 9 [15]	SLA glass/polyester laminate	Polyester	$R = 0.1$	-0.1500	6.67	Secondary only; not representative of glider epoxy structures
SAND2010-7052, Table 9 [15]	SLB glass/polyester laminate	Polyester	$R = 0.1$	-0.1900	5.26	Secondary only; lower-performance non-epoxy system, not used for selection
SAND2010-7052, Table 9 [15]	SLC glass/polyester laminate	Polyester	$R = 0.1$	-0.1500	6.67	Secondary only; not representative of glider epoxy structures
AIAA 2012 / SNL-MSU-DOE database, Table 7 [16]	Multidirectional glass/vinyl ester laminate $[(\pm 45)_2/(0)_2]_s$	Vinyl ester	$R = 0.1$	-0.146	6.85	Secondary only; not representative of glider epoxy structures
AIAA 2012 / SNL-MSU-DOE database, Table 7 [16]	Multidirectional glass/polyester laminate $[(\pm 45)_2/(0)_2]_s$	Polyester	$R = 0.1$	-0.151	6.62	Secondary only; not representative of glider epoxy structures

The data show that composite fatigue exponents are not universal material constants. They depend on fibre type, matrix, layup, fibre volume, loading direction, stress ratio, manufacturing process, adhesive/detail geometry and failure mode. General composite fatigue references also emphasize this dependence and the need to treat S/N parameters as system-specific rather than universal material properties [19] [20] [21].

For the purpose of a generic ageing-glider Structural Life Monitoring method, however, the goal is not to claim a measured material property for each glider type. The goal is to select a conservative bounding exponent for comparative EFI assessment when type-specific laminate data are not available. Within the reviewed epoxy-based GFRP/CFRP data, the steepest identified GFRP epoxy stress-fit exponent is $B = -0.1556$ for the QQ1 glass/epoxy multidirectional laminate at $R = 0.1$. Other epoxy GFRP values are less severe, and carbon-dominated epoxy laminates show much shallower S/N slopes and therefore lower fatigue sensitivity.

Therefore, to avoid taking into account each glider's specific materials and construction, the selected generic value is:

$$b = -0.16$$

and therefore:

$$K = -\frac{1}{b} = 6.25$$

This value is slightly steeper than the steepest identified GFRP epoxy value in the reviewed SNL/MSU/DOE data and is strongly conservative for CFRP-dominated epoxy structures. Vinyl ester and polyester data are not used to define the value because they are not representative of composite glider epoxy structures. They are retained only as a secondary comparison showing that the selected value is also on the conservative side of several lower-performance non-epoxy multidirectional glass/polymer laminates.

The selected value is particularly conservative when compared with axial or fibre-dominated laminates. This distinction is important because the highest cyclic stresses in conventional composite gliders are primarily associated with wing bending and are carried by spar caps, spar flanges and other primary load-carrying elements whose fibres are predominantly aligned with the principal stress direction. These regions are closer to axial or axial-dominated laminate behaviour than to strongly off-axis or resin-dominated coupon behaviour.

In the SNL/MSU/DOE data, axial or axial-dominated glass/epoxy laminates show much shallower stress-life exponents than the selected value. For example, an epoxy UD fabric laminate reports $B = -0.088$, corresponding to $K = 11.4$, while the selected SLM value $b = -0.16$ corresponds to $K = 6.25$. Carbon-dominated epoxy axial laminates are even less fatigue-sensitive, with reported values such as $B = -0.0226$, corresponding to $K = 44.25$. Therefore, the selected value $b = -0.16$ is very conservative relative to the fibre-aligned load paths expected to dominate global wing-bending fatigue in composite gliders.

The selected value $b = -0.16$ is also conservative relative to the reviewed off-axis, multidirectional and structural-detail fatigue data in the SNL/MSU/DOE database. Reported examples include biax glass/epoxy $B = -0.092$, transverse glass/epoxy $B = -0.114$, ply-drop delamination details between $B = -0.099$ and $B = -0.120$, adhesive lap-shear details around $B = -0.109$ to $B = -0.135$, and triax glass/epoxy $B = -0.143$. These values are all less severe than the selected $b = -0.16$. Therefore, the selected exponent remains conservative not only for axial fibre-dominated load paths, but also for the reviewed off-axis and structural-detail data. Nevertheless, the EFI remains a load-factor-based fatigue index and does not replace inspection of local damage, repairs, impact damage, ageing effects or airworthiness-significant defects.

The selected value should be treated as a conservative engineering parameter for the SLM EFI, not as a universal composite-material property.

The equation for EFI accumulation for the *real* spectrum becomes:

$$EFI_{real} = \frac{1}{C'} \sum_i n_{i,real} |LF_{i,real} - LF_{average,real}|^{6.25}$$

$$\text{where } C' = 3 \sum_i n_{i,ref} |LF_{i,ref} - LF_{average,ref}|^{6.25}$$

As long as EFI_{real} stays below (and the required inspections and continued-airworthiness conditions are satisfied), the glider can continue operations.

The equation for the life extension becomes:

$$t_{\text{extension}} = t_{\text{remaining ref}} \times \left(\frac{\sum_i \lfloor \frac{n_{i,ref} \times t_{real}}{6000} \rfloor \times |LF_i - LF_{average,ref}|^{6.25}}{\sum_i n_{i,real} \times |LF_i - LF_{average,real}|^{6.25}} - 1 \right)$$

From a fatigue standpoint, this is a conservative approach because the selected exponent is slightly steeper than the steep end of reviewed GFRP epoxy laminate fatigue data and strongly penalizes the EFI calculation relative to CFRP-dominated epoxy structures.

3.3.4 Interpretation of the Equivalent Fatigue Index

The EFI is an engineering fatigue-consumption index constructed to compare the measured operational load spectrum with the accepted certification reference spectrum. This distinction is important but does not weaken the method: the certification decision is based on preserving the approved fatigue exposure budget, not on predicting every microscopic damage mechanism in the composite structure.

The EFI is conservative by construction for two main reasons. First, the certification reference condition treats the approved service life, multiplied by the applicable safety factor, as the fatigue budget to be consumed. In reality, many fatigue substantiations are completed without failure or significant observable damage, but any unobserved residual margin is not credited. The SLM method therefore conservatively assumes that the certification reference exposure corresponds exactly to the allowable limit, $EFI = \frac{1}{3}$ at 12,000 hours. Second, the Palmgren-Miner/Basquin calculation uses the accepted reference spectrum, the same counting method for reference and measured spectra, and the conservative bounding exponent $b = -0.16$ selected above.

Under these assumptions, the EFI is expected to overestimate fatigue consumption associated with normal in-envelope flight loading for a sound structure. This is particularly true for the primary wing-bending load paths, where the most highly loaded fibres are predominantly aligned with the principal stress direction and where the selected exponent is much more severe than the reviewed axial GFRP/CFRP epoxy fatigue data.

Continued operation beyond the original life limit is therefore justified when both conditions are satisfied:

$$EFI_{total} < \frac{1}{3}$$

and the aircraft remains in acceptable structural condition through the applicable inspection and continued-airworthiness programme. This interpretation preserves the safety level associated with the original Type Certificate while replacing assumed reference-spectrum usage by measured real-spectrum usage.

4. Results and discussion

4.1 Comparative occurrences analysis

Preliminary calculations performed on a Janus B glider, using approximately 35 hours of flight data in various conditions, indicate that the real load spectrum produces significantly lower load factor occurrences than the reference model predicts. This reduction in load factor occurrences also translates into a lower cumulative EFI. The corresponding EFI ratio supports the potential for life extension.

The load spectrum used as a reference is the Kossira & Reinke flight-only spectrum, which excludes simple aerobatics and ground operations. This represents the most conservative choice within the available Kossira & Reinke spectra: adding simple aerobatics and ground operations would introduce additional occurrences at higher load factors, increasing the reference EFI accumulation and yielding a more favorable EFI ratio. The life extension results presented here therefore constitute a conservative lower bound of the potential gain.

4.2 Palmgren-Miner EFI and ratio evaluation

The simplified ratio:

$$R = \frac{\sum_i \lfloor \frac{n_{i,ref} \times t_{real}}{6000} \rfloor \times |LF_i - LF_{average,ref}|^{6.25}}{\sum_i n_{i,real} \times |LF_i - LF_{average,real}|^{6.25}}$$

provides a direct method to evaluate the difference in EFI accumulation. Since real flight loads are less severe than those predicted by the Kossira & Reinke distribution, R is expected to be significantly greater than one, thereby justifying an extension in the glider's operational life.

The following is an example of calculations performed with the conservative value of $-1/b = 6.25$, corresponding to the selected conservative GFRP/CFRP epoxy bounding value.

A very preliminary analysis has been performed on several flights with different pilots on a Janus B. The recording performed at 20 Hz sample rate was processed using the Kossira & Reinke counting method.

Using the Janus B data and the Kossira & Reinke reference spectrum, Table 2 provides the different steps to compute the EFI ratio. Because of the limited data, this table should be considered more as an illustration of the method.

		Kossira occurrences normalized @ 6000FH				Janus occurrences normalized @ 6000FH			
		↓				↓			
		Damage ratio =				26			
		S/N slope = K =				6.25			
		avg n Kossira (g)=				1.10			
		avg n JANUS (g) =				1.07			
		class #	n (g)	Kossira 6000 FH	n - avg Kossira * K	Kossira: occ * n - avg * K	JANUS 6000 FH	n - avg JANUS * K	JANUS: occ * n - avg * K
		1	-3.84375	0.0	21834.52802843	0.00	0.0	20988.74408562	0.00
		2	-3.53125	0.0	14520.83715246	0.00	0.0	13921.09049055	0.00
		3	-3.21875	0.0	9385.67077614	0.00	0.0	8970.51411678	0.00
		4	-2.90625	0.0	5870.90987604	0.00	0.0	5591.38933572	0.00
		5	-2.59375	0.0	3535.07946904	0.00	0.0	3352.85427818	0.00
		6	-2.28125	6.2	2035.26009868	12597.62	0.0	1920.89125885	0.00
		7	-1.96875	46.4	1110.67862153	51481.32	0.0	1042.08321970	0.00
		8	-1.65625	281.9	567.94447933	160094.80	0.0	529.01216981	0.00
		9	-1.34375	1165.0	267.89541020	312096.63	0.0	247.26333330	0.00
		10	-1.03125	2988.8	114.01394111	340766.72	0.0	103.99606237	0.00
		11	-0.71875	6098.6	42.37290081	258414.11	0.0	38.03941225	0.00
		12	-0.40625	10800.3	13.06443217	141100.10	352.9	11.46643757	4046.98
		13	-0.09375	21279.8	3.06231479	65165.51	7058.8	2.59649207	18328.18
		14	0.21875	133934.4	0.46142620	61800.84	35294.1	0.36866753	13011.80
		15	0.53125	1423065.8	0.03016180	42922.23	352941.2	0.02123916	7496.17
		16	0.84375	6624972.4	0.00021328	1412.95	5647058.8	0.00009558	539.74
		17	1.15625	6196570.8	0.00000001	0.07	6024705.9	0.00000020	1.23
		18	1.46875	4268809.2	0.00188240	8035.60	1852941.2	0.00313543	5809.76
		19	1.78125	728422.6	0.08887528	64738.76	264705.9	0.11765512	31144.00
		20	2.09375	149720.0	0.94744078	141850.82	42352.9	1.14965269	48691.17
		21	2.40625	49805.3	5.25138689	261546.75	17647.1	6.08679718	107414.07
		22	2.71875	24545.5	20.11073894	493627.72	5294.1	22.66026492	119966.11
		23	3.03125	15081.0	60.69932142	915403.54	529.4	67.09491690	35520.84
		24	3.34375	10346.1	155.14737696	1605175.46	0.0	169.13217557	0.00
		25	3.65625	7025.2	350.77035935	2464238.36	0.0	378.39556253	0.00
		26	3.96875	3133.2	721.69502378	2261235.39	0.0	772.15919982	0.00
		27	4.28125	449.9	1377.93673853	619944.26	0.0	1464.57258831	0.00
		28	4.59375	32.7	2475.97647357	81087.50	0.0	2617.38964309	0.00
		29	4.90625	0.0	4230.88160227	0.00	0.0	4452.24574008	0.00
		30	5.21875	0.0	6930.01114231	0.00	0.0	7264.52308522	0.00
		31	5.53125	0.0	10948.34314394	0.00	0.0	11438.84184967	0.00
		32	5.84375	0.0	16765.45946373	0.00	0.0	17466.21208361	0.00
		↑							
		Weighting factor centered on Kossira average g due to S/N curve				Weighting factor centered on Janus average g due to S/N curve			

Table 2: Palmgren-Miner ratio and life extension calculation

In this table the parameter $K = -1/b$, corresponds to the exponent of the S/N curve and the Basquin parameter.

The EFI ratio is simply the ratio of the $Kossira\ occ * |n - avg|^K$ column and $JANUS\ occ * |n - avg|^K$ column sum.

where

$Kossira\ occ * |n - avg|^K$ corresponds to $\sum_i \left[\frac{n_{i,ref} \times t_{real}}{6000} \right] \times |LF_i - LF_{average,ref}|^{6.25}$

$JANUS\ occ * |n - avg|^K$ corresponds to $\sum_i n_{i,real} \times |LF_i - LF_{average,real}|^{6.25}$

As shown in Table 2, the ratio between the Kossira & Reinke reference EFI and the real-flight EFI is 26. Assuming the glider continues to operate with a similar load spectrum until it reaches 12,000 hours (i.e. in ~300 hours), the life extension would be around 7,500 hours ((26-1) x 300).

Table 2 can also be used to help understand the sensitivity of the Palmgren-Miner ratio R to the type of flight operations. If we imagine a pilot performing a constant 4g loop every two hours of flying, because the Table 2 data is normalized to 6,000 hours, this would add 3,000 occurrences to classes 17 to 26 and would reduce the R ratio example from 26 down to 2.

This illustrates that the proposed method will justify life extension according to the use of the glider, and that a few high-load maneuvers will have a significant effect on the R ratio, as per the Palmgren-Miner rule. It should be noted that the R ratio of 2, due to the loops, is not accurate since the reference spectrum used for calculations is a flight-only, without aerobatics, spectrum. The purpose of the illustration is only to show that the R ratio obtained when flying normally would be reduced by a factor of 13 due to aggressive maneuvers.

Key assumptions and limitations

The proposed SLM method is based on the following assumptions and limitations:

- SLM does not reopen the Type Certificate structural substantiation.
- SLM does not determine the absolute physical damage state or absolute fatigue life of the composite structure; the method is based on a global reference-spectrum comparison rather than a local failure-mode prediction.
- SLM uses the approved certification life exposure as the conservative reference limit.
- SLM compares measured operational severity against this accepted reference severity.
- The Palmgren-Miner calculation is used as a common comparison framework for both spectra.
- The Basquin parameter is used as a conservative severity-weighting parameter, applied consistently to both the reference and measured spectra.
- The method remains dependent on continued inspections, especially for defects, repairs, bonds, metallic fittings, and unforeseen local deterioration.
- Missing or invalid flight data must be replaced by conservative reference-spectrum usage.

This study focuses on composite glider parts, hence the conservative GFRP/CFRP epoxy 6.25 Basquin parameter. It is however interesting to note that, if an aluminum-typical Basquin parameter, $-1/b = 5$, is used, it results in a significant life-extension ratio of 12. This could be used to estimate the metallic parts life extension which would be around 3,600 hours (12 x 300), based on the Janus data. This is significantly more than the repetitive 1,000-hour inspections, currently required after 9,000 hours. This legitimizes the continuation of these recurring inspections at 13,000 hours, 14,000 hours and beyond.

It should also be noted that, contrary to the composite structures, these metallic parts (wing shear pins and bushings, spar pins and bushings, tail fittings, etc.) can generally be inspected and eventually replaced during the repetitive inspections, reducing further the risk of failure.

4.3 Practical implications, safety and certification

Using this novel Structural Life Monitoring method, it is possible to tailor a given glider's life, beyond the initially certified 12,000-hour limit, as a function of its specific usage. This life limit extension is achieved while maintaining the level of equivalent fatigue exposure accepted during certification.

EASA has already accepted several methods to comply with fatigue strength regulation requirements. This

legitimizes SLM as an additional alternative solution for aging gliders.

EASA Certification Memorandum CM-S-006 endorses the use of the Kossira & Reinke spectra which have been designed to cover all types of usage, including some operations which are rarely performed by most gliders. While the proposed solution is consistent with these EASA reference load spectra and maintains significant calculation margins, we still anticipate a large difference between the measured and the reference EFI accumulations. This has been confirmed during initial tests on several instrumented gliders. Still, the exact extension granted by the proposed SLM method will account for actual glider use, on a serial number basis.

As the life increase is proportional to the time the real spectrum is observed, it is advisable to instrument gliders as soon as possible and before they reach 12,000 hours, to be able to gain significant life extension. Early life instrumentation might not be necessary when solely considering aging glider life extension, but would provide valuable data and statistics on glider usage.

4.3.1 Concept of Operation and Deployment Proposals

4.3.1.1 Concept of Operation Proposal

The concept of operation could be divided into 3 main phases:

Phase 1 — Instrumentation and Data Collection (below Type Certificate safe life limit)

This phase covers the period during which the SLM recorder is installed, and the glider has not yet reached its safe life limit. It includes:

- Sensor installation in accordance with CS-STAN 104b / Form 123.
- Sensor installation calibration with the glider in level-flight attitude and wings level, so that the recorder-corrected vertical axis is aligned with the gravity vector in the reference flight attitude.
- Annual 6-face calibration during the annual inspection.
- Compliance with Part ML; installation and annual calibration results recorded in the aircraft logbook.
- Missing data policy: if a flight occurs without valid SLM data (sensor fault, data corruption, missed calibration), EFI for that flight is substituted at the reference Kossira & Reinke EFI rate for the flight duration.
- Flight data uploaded to the central web repository; post-processing tools applied to guarantee data integrity.

Phase 2 — Fatigue Evaluation when reaching Type Certificate safe life limit

This phase includes:

- a dedicated fatigue inspection of the aircraft condition (equivalent in scope to the existing recurring 1,000-hour inspections).
- an assessment of the cumulative EFI to possibly permit operation up to the originally certified fatigue reference limit.
- inspection beyond the composite structure, as well as an overall assessment of the glider's condition (incidents/repairs).

Phase 3 — Operation Beyond Type Certificate safe life limit

This phase is a continuation of Phase 1 monitoring operations, with the following additional provisions:

- Go / No-Go decision at each Airworthiness Review Certificate (ARC) renewal, based on the SLM equivalent fatigue index report and the associated structural inspection status.
- Go criterion: cumulative $EFI_{total} < 1/3$, using the approved reference assumptions, AND the current EFI rate confirms that $EFI = 1/3$ will not be reached before the next scheduled inspection.
- Recurring structural inspection continuing from Phase 2, with a maximum interval of 1,000 hours or 5 years, whichever is sooner.
- This process continues until the aircraft has exhausted its remaining EFI potential and is retired from service

4.3.1.2 Validation of the concept and Deployment proposal

The SLM solution has never been implemented on gliders and there is no Interpretative Material (IM) or Acceptable Means of Compliance (AMC) to support CS-22 certification.

One important step toward SLM approval will be the publication of a Certification Review Item, which shall support the establishment of IM or AMC.

The CRI shall cover acceptable:

- fatigue/EFI calculation methods
- reference load spectrum and composite material parameter
- onboard recording solution
- postprocessing solution
- CONOPS and Instructions for Continued Airworthiness (including impact on AFM/AMM/AMP)
- limits of proposed methodology and SLM applicability

As part of the validation activity, it may also be useful to review available certification-test history for composite gliders, where such information can be accessed. Information on whether previous full-scale fatigue tests completed the required spectrum without significant damage, or whether specific damage modes were observed, could help improve the understanding of actual structural margins. This information should not be considered a prerequisite for the initial SLM approach, which remains deliberately conservative. However, it could support later refinement of the method and may allow some conservative assumptions to be reviewed, provided the evidence supports such refinement and the certification authority accepts it.

An EASA STC could be used to develop the required certification justification documents.

The STC could be applicable to a small group of gliders in experimentation. They would provide flight data as well as validate the CONOPS.

The STC would produce the following documents which will be required to get the SLM approved:

- Certification Program
- Certification Review Item
- Methodology Report
- Instructions for Continued Airworthiness
- Installation, Operating and Maintenance Manual

Once the STC is approved, the gliders in experimentation would be allowed to continue their operation using SLM.

The STC documents could be made available to the interested Type Certificate holders. They could develop modifications and Service Bulletins to permit SLM operation on selected glider types.

5. Conclusion

This paper presents a methodology for extending the operational life of composite gliders using Structural Life Monitoring. By measuring in-flight accelerations and processing them with the Palmgren-Miner rule, it is possible to compute an Equivalent Fatigue Index for an individual glider. This index should not be interpreted as a direct measurement of physical composite damage, but as a conservative usage-severity index calculated consistently for both the reference certification spectrum and the measured operational spectrum. Crucially, both the reference and real flight spectra are processed using the same Kossira & Reinke counting method, ensuring a consistent and legitimate comparison. As long as the cumulative EFI remains below the certification reference limit, and the required inspections and continued-airworthiness provisions are satisfied, continued operation may be justified — without any requirement to reopen Type Certificate documents or revise certified structural limits.

Preliminary results show that, even when applying conservative calculation methods and worst-case composite material fatigue parameters, EFI accumulation from actual flights is orders of magnitude lower than EFI accumulation predicted by the reference spectrum. An EFI ratio of around 26 has been observed on an instrumented glider, corresponding to potential life extensions of several thousand hours for typical training and cross-country operations. Interestingly, the same approach applied to metallic components, using aluminum Basquin parameters, also yields significant life extension, legitimizing the continuation of recurring structural inspections beyond the current life limit. It should however be noted that life extension is directly dependent on how the glider is flown — gentler flying results in proportionally greater extension.

The Basquin parameter used in the EFI calculation is not claimed to represent the exact failure mode of the structure. It is used as a conservative severity-weighting parameter applied consistently to both the reference and measured spectra. The purpose of SLM is therefore not to predict the absolute fatigue life of a composite structure from first principles, but to compare measured operational severity with the accepted certification reference severity.

SLM life extension opportunities are consistent with life assessments performed on similar composite structures in other industries, and with the life assessment performed by TRAFI for the PIK-20D in 2015.

The proposed method is conservative by design, compatible with EASA fatigue strength regulation requirements, and provides economic and environmental benefits while maintaining the same safety level as the original Type Certificate (same certified reference limit as the original TC). A structured three-phase Concept of Operation — covering instrumentation, fatigue evaluation at the certified life limit, and monitored operation beyond it — has been outlined as a practical deployment framework. It is advisable to first validate this approach on a selected fleet of instrumented gliders. The findings could then support the establishment of an EASA Certification Review Item, leading to an Acceptable Means of Compliance for SLM under CS-22, and ultimately deployed on aging gliders through Supplemental Type Certificates or manufacturer Service Bulletins.

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